🔒 Data Leakage —

When Your Model Khows What It Should’n’t

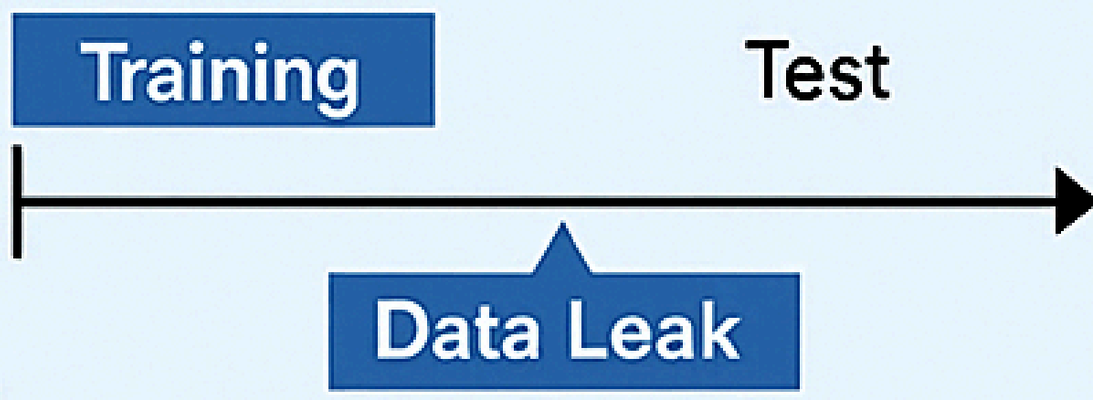
Imagine going into an exam, and the teacher accidentally prints the answer key on the back of your test. You ace it, but you really earn that grade?

That's what happens with data leakage — your model “cheats” by learning from information it should never have seen.

🧠 What is Data Leakage?

Data leakage is when your model gets access to information during training that it won’t have in the real world

So, the model ends up learning the wrong patterns, usually ones that are too good to true — because, well, they are.



🚨 How Does Leakage Happen

- You might think, “I’d never let that happen!

But data leakage is sneeky.
It hides in the shadows.
of preocesssing, feature pineering, and poor data splits

Here are som common types:

1. **Train–test contamination**
   You accidentally let test data slip into your training set
   - “We just shuffied everything together, no big dea!”
2. **Target leakage through features**

🚨 Red Flags — Signs of Data Leaıakage

🕶Do This	Non’t Do This
Split training and test strictly	Mix data before separding
Remove features, too close to targat	Include “tuture knowledge” feature
Use chronological splits in time sires	Randomly shuff time data

🎓 My Take on It…

For me, data leakage is like givin your model the answer key befo the test.